

Extending Equilibrium Matching: Second-Order Dynamics, Latent Spaces, and Conditional Generation

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Equilibrium Matching

Problem Statement

Current SOTA generative models (Diffusion, Flow Matching) rely on **non-equilibrium dynamics** (time-dependent vector fields). This enforces fixed integration horizons and prevents flexible optimization-based sampling.

Research Goal

Develop extensions to Equilibrium Matching that enhance inference speed, training efficiency, and stability while rigorously preserving its inherent time-invariant vector field properties.

Tasks

- ▶ Extend EqM to conditional generation and latent spaces.
- ▶ Regularize the training using Second-Order (Hessian) dynamics.
- ▶ Analyze the sampling trajectory using UMAP.
- ▶ Analyze the stability and geometric properties of EqM.

Baseline: Equilibrium Matching

We define a corruption process via interpolation $\gamma \sim U[0, 1]$:

$$\mathbf{x}_\gamma = \gamma \mathbf{x} + (1 - \gamma) \boldsymbol{\epsilon}$$

Unlike Flow Matching, EqM minimizes the difference between the model output $f(\mathbf{x}_\gamma)$ and a target gradient field pointing from noise to data:

$$\mathcal{L}_{\text{EqM}} = \mathbb{E}_{\mathbf{x}, \boldsymbol{\epsilon}, \gamma} \left[\left\| f(\mathbf{x}_\gamma) - \underbrace{(\boldsymbol{\epsilon} - \mathbf{x})c(\gamma)}_{\text{Target Gradient}} \right\|^2 \right]$$

- ▶ **Implicit Energy:** $f(\mathbf{x}) \approx \nabla_{\mathbf{x}} E(\mathbf{x})$.
- ▶ **Optimization Sampling:** Sampling is performed via Gradient Descent (GD) rather than ODE integration.

Proposed Extension: Second-Order Regularization

Standard EqM penalizes the first derivative but ignores curvature (Hessian). We introduce a regularization term to align the curvature of the energy landscape.

Method: Using Hutchinson's Estimator for Jacobian-Vector Products (JVP). Let $\mathbf{v} \sim \mathcal{N}(0, I)$.

$$\mathcal{L}_{2\text{nd}} = \left\| \underbrace{(\nabla_{\mathbf{x}_\gamma} f(\mathbf{x}_\gamma)) \cdot \mathbf{v}}_{\text{Predicted Curvature}} - \underbrace{(-c(\gamma)\mathbf{v})}_{\text{Target Curvature}} \right\|^2$$

Hypothesis

Explicitly controlling the Hessian eigenvalues ensures the energy landscape is convex near the data manifold, improving convergence of Newton/NAG methods.

Proposed Extension: Conditional Generation

To enable class-conditional generation (e.g., MNIST/CIFAR classes), we introduce Adaptive Group Normalization (**AdaGN**).

Given class embedding $\mathbf{e}_y$:

$$\text{AdaGN}(\mathbf{h}, y) = \text{GroupNorm}(\mathbf{h}) \cdot (1 + \mathbf{s}(\mathbf{e}_y)) + \mathbf{b}(\mathbf{e}_y)$$

This modulates the energy landscape:

$$E(\mathbf{x}, y) \neq E(\mathbf{x})$$

effectively creating distinct "points of attraction" for different classes within the same vector field.

Proposed Extension: Importance Sampling & Smoothing

1. **Smooth Gradient Magnitude** $c(\gamma)$: Original EqM uses piecewise linear functions. We propose a smooth approximation to ensure differentiability of the target field:

$$c_{\text{smooth}}(\gamma) = \frac{r(\gamma)}{\sqrt{1 + r(\gamma)^2}}, \quad r(\gamma) = \frac{1 - \gamma}{1 - a}$$

2. **Gamma Scheduling**: Uniform sampling $\gamma \sim U[0, 1]$ is probably is not the most efficient. We investigate Beta Distribution sampling $\gamma \sim \text{Beta}(\alpha, \beta)$ to prioritize the "confusion zone" ($\gamma \approx 0.5$) where the gradient direction is most critical.

Experiments: Sampling Dynamics

We compared three sampling strategies on MNIST/CIFAR-10:

1. **Gradient Descent (GD)**: Baseline.
2. **Nesterov Accelerated Gradient (NAG)**: Momentum-based.
3. **Muon**: Momentum orthogonalization.

Adaptive Compute: We stop sampling when $\|\nabla E(\mathbf{x})\| < \epsilon$.

- ▶ Simple digits ('1'): ~ 20 steps.
- ▶ Complex digits ('8'): ~ 45 steps.



Figure: Samples NAG

Analysis: UMAP Trajectory Visualization

To understand the optimization landscape, we projected the sampling trajectory $\mathbf{x}_0 \rightarrow \mathbf{x}_K$ into 2D using UMAP.

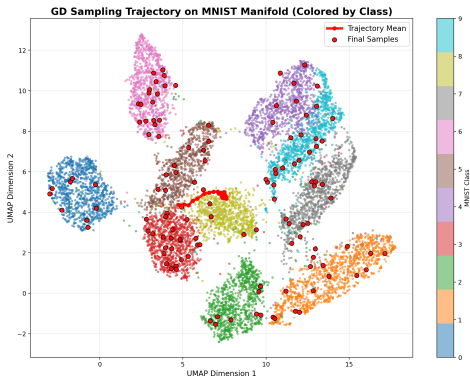


Figure: UMAP visualization

Observation: The trajectory is non-linear, following the manifold curvature rather than a straight line, confirming the learned energy landscape structure.

Conclusions

1. **Conditional Generation:** AdaGN successfully modulates the equilibrium points, allowing controllable generation without classifier guidance.
2. **Sampling:** NAG outperforms standard GD. Naive Muon implementation requires further refinement for generative tasks.

Next Steps

- ▶ Implement ICNN (Input Convex Neural Networks) to guarantee global energy convergence.
- ▶ Further develop ways to generate explicit energy landscapes from known gradient field
- ▶ Make EqM scaling to Latent Diffusion (LEqM) contexts stable.
- ▶ Stabilize second-order vector field to avoid over-constraining the model.